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324615 (24)

BE (6th Semester)

Examination, April-May, 2014

Branch : Electrical, EEE

DIGITAL SIGNEL PROCESSING

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

te : Part (a) is compulsory and answer any two parts from b, c and d.

1. (a) Determine whether the signal is periodic if fundamental period of x_1 is 0.133 sec & fundamental period of x_2 is 0.1 sec. 2

$x(t) = x_1(t) + x_2(t)$

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(b) With example explain shifting, folding and time scaling operations on discrete time signals. 7

(c) Explain the terms single-sided spectrum and double sided spectrum with respect to a signal. Also sketch the single and double sided amplitude and phase spectra of signal $x(t) = 8 \sin (20\pi t - \pi/6), -\infty < t < \infty$. 7

(d) Explain the terms time invariant and time variant system. Determine whether system given below is linear and non-linear as well as time variant and time invariant system 7

$y[n] = x[n] + nx[n + 1]$

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Q. 2. (a) Check the stability of filter for 2

$$H(z) = \frac{z^2 - z + 1}{z^2 - z + 1/2}$$

(b) Find the impulse response for the causal system $y(n) - y(n-1) = x(n) + x(n-1)$. Also find the response of the system to inputs $x(n) = u(n)$ and $x(n) = 2^{-n} u(n)$. Test its stability. 7

(c) Find the impulse response, frequency response, magnitude and phase response of second order system : 7

$$y(n) - y(n-1) + \frac{3}{16} y(n-2) = x(n) - \frac{1}{2} x(n-1)$$

(d) A second order discrete time system is

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characterised by the difference equation of

$y(n) - 0.1\, y(n-1) - 0.02\, y(n-2) = 2x(n) - x(n-1)$. Determine $y(n)$ for $n \geq 0$ when $x(n) = u(n)$ and the initial conditions are $y(-1) = -10$ & $y(-2) = 5$. 7

Q. 3. (a) What are the properties of DFT. 2

(b) Compute the convolution $y(n) = x(n) * h(n)$ of signal $x(n) = \{1, 1, \underset{\uparrow}{0}, 1, 1\}$ and $h(n) = \{1, -2, \underset{\uparrow}{-3}, 4\}$. 7

(c) Find the inverse DFT of $X(k) = \{1, 2, 3, 4\}$. 7

(d) Given $x(n) = 2^n$ and $N = 8$, find $X(k)$ using DIT FFT algorithm. 7

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Q. 4. (a) What is an IIR filter? Compare its characterisitc with an FIR filter. 2

(b) Determine H(z) using the impulse invariant technique for analog system function : 7

$$H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

(c) A digital filter with 3 dB bandwidth of 0.25π is to be designed from the analog filter whose system is $H(s) = \frac{\Omega_c}{s+\Omega_c}$ use bilinear transformation & obtain H(z). 7

(d) Design a digital Chebyshev filter to satisfy the constraints : 7

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$$0.707 \le |H(e^{jw})| \le 1 \quad 0 \le w \le 0.2\pi$$

$$|H(e^{jw})| \le 0.1 \quad 0.5\pi \le w \le \pi$$

using bilinear transformation and assume T = 1 sec.

Q. 5. (a) What do you mean by parallel realisation of IIR system. 2

(b) Determine direct forms I and II for the second order filter given by : 7

$$y(n) = 2b \cos w_0 y(n-1) - b^2 y(n-2) + x(n) - b \cos w_0 x(n-1)$$

(c) (i) Realise using ladder structure :

$$H(z) = \frac{2+8z^{-1}+6z^{-2}}{1+8z^{-1}+12z^{-2}}$$

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(ii) Realise a system defined by following state space :

$$\begin{bmatrix} r_1(n+1) \\ r_2(n+2) \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} + \begin{bmatrix} 4 \\ 6 \end{bmatrix} x(n)$$

$$y(n) = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix}$$ 7

(d) Explain various stages used in video processing also explain how speech signal is generated ? 7

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